

Measurement-Induced Entanglement Phase Transitions in 1D Random Clifford Circuits

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We study finite-size signatures of measurement-induced entanglement transitions in one-dimensional random Clifford circuits using an Aaronson-Gottesman stabilizer tableau simulator implemented from scratch. For $L \in \{8, 12, 16, 20, 24\}$, 200 disorder realizations per parameter point, and single-site measurement rates $p \in [0, 0.5]$, we observe the expected crossover from volume-law entanglement at low measurement rate to area-law entanglement at high measurement rate. Adjacent-size crossings and finite-size scaling fits give effective critical rates $p_c^{\text{eff}} \approx 0.20$, above the thermodynamic literature value $p_c \approx 0.16$ [1], consistent with well-documented finite-size corrections at these small sizes. A log-scaling fit near the literature critical point gives a finite-size coefficient $\alpha_{\text{eff}} \approx 0.27$, which shows the expected logarithmic form but does not quantitatively reproduce the asymptotic literature value [1]. We additionally measure the full entanglement profile $S(A)$ as a function of subsystem size $|A|$ for representative values of p , directly resolving the linear volume-law growth, the approximately flat area-law behavior, and the intermediate sublinear growth near the critical region. The project reproduces the qualitative phenomenology of the MIPT and highlights the limitations of extracting universal exponents from $L \leq 24$ data.

INTRODUCTION

At the heart of quantum dynamics lies a fundamental competition. Unitary Clifford evolution drives generic quantum states toward the volume law, half-chain entropy $S(L/2) \propto L$, at an extensive rate set by the circuit. Projective measurements do the opposite—they collapse superpositions, extracting information from the system and suppressing quantum correlations. The measurement rate p , the fraction of qubits measured per layer, controls the balance between these two tendencies. A natural question emerges: is there a sharp boundary in p separating the entangled and disentangled regimes, or does the entropy decrease smoothly?

The answer turns out to be dramatic. Volume-law entanglement, $S(L/2) \propto L$, is characteristic of quantum-chaotic states and arises in random quantum circuits with sufficient entangling power [1]. Area-law entanglement, $S(L/2) = O(1)$, is independent of system size and characterizes both ground states of gapped one-dimensional Hamiltonians and classically simulable stabilizer product states. These two behaviors are not merely different points on a continuum; they represent qualitatively distinct phases of quantum matter with different information-theoretic and computational properties.

The discovery that hybrid quantum circuits exhibit a sharp transition between these phases came in 2018–2019 from two independent groups. Li, Chen, and Fisher introduced and studied hybrid quantum circuits, showing that projective measurements induce a genuine phase transition with universal critical exponents [1, 2]. Simultaneously, Skinner, Ruhman, and Nahum demonstrated the same transition and connected it to random tensor networks and an effective critical field theory [3]. The key surprise is that the transition is not merely a smooth

crossover: it is a *sharp*, system-size-scaling critical point governed by universal exponents p_c and ν in the thermodynamic limit. The physical mechanism at large p is the quantum Zeno effect: sufficiently frequent measurement freezes the quantum state into a product state, completely suppressing scrambling.

Clifford circuits provide the ideal tractable substrate for this investigation. The Gottesman-Knill theorem guarantees that Clifford circuits—generated by Hadamard H , phase S , and CNOT gates—can be simulated efficiently on a classical computer [4]. The Aaronson-Gottesman stabilizer tableau [5] implements this simulation in $O(n)$ per gate and $O(n^2)$ per measurement, reducing the exponentially hard quantum simulation problem to linear algebra over $\text{GF}(2)$. This enables us to access system sizes ($L \leq 24$) and disorder averages (200 samples per point) sufficient to resolve the qualitative finite-size signature of the phase transition.

In this paper we implement the full Aaronson-Gottesman tableau from scratch and use it to (i) map out the phase diagram $S(L/2)$ vs. p for $L \in \{8, 12, 16, 20, 24\}$; (ii) measure the full entanglement profile $S(A)$ vs. subsystem size $|A|$ for three representative values of p , directly resolving the linear and flat profiles of the two phases; (iii) measure a log-scaling coefficient at $p = 0.16$; (iv) perform a qualitative finite-size scaling collapse; and (v) discuss connections to quantum error correction and threshold phenomena in machine learning. The paper is organized as follows: Sec. reviews the theoretical framework; Sec. describes the circuit model and simulation protocol; Sec. presents numerical results including the entanglement profile; Sec. discusses physical implications; and Sec. concludes.

BACKGROUND

Stabilizer States and Clifford Circuits

Clifford circuits are generated by three gates: Hadamard H , phase S , and CNOT. Their defining property is that they map Pauli operators to Pauli operators under conjugation:

$$\begin{aligned} HXH^\dagger &= Z, & HZH^\dagger &= X, \\ SXS^\dagger &= Y, & SZS^\dagger &= Z, \\ \text{CNOT}(X \otimes I) \text{CNOT}^\dagger &= X \otimes X. \end{aligned} \quad (1)$$

A *stabilizer state* is any state that can be obtained from $|0\rangle^{\otimes n}$ by applying a Clifford circuit. Such a state is uniquely specified by its *stabilizer group*: the abelian subgroup of n -qubit Paulis $\{g_i\}$ for which $g_i|\psi\rangle = |\psi\rangle$. The Gottesman-Knill theorem guarantees efficient classical simulation [4].

We use the Aaronson-Gottesman tableau [5]: a $(2n) \times (2n+1)$ binary matrix over $\text{GF}(2)$ encoding n destabilizer generators (rows $0, \dots, n-1$) and n stabilizer generators (rows $n, \dots, 2n-1$), with columns $0, \dots, n-1$ encoding the X -part of each generator, columns $n, \dots, 2n-1$ encoding the Z -part, and column $2n$ encoding the phase bit ($0 \rightarrow +1, 1 \rightarrow -1$). Single-qubit gates update the tableau in $O(n)$ time (all rows simultaneously via NumPy vectorization); measurement is $O(n^2)$. The full gate update rules are given in Appendix .

Entanglement Entropy from the Stabilizer Tableau

For a stabilizer state of n qubits with bipartition $A \cup B$ (B the complement of A), the von Neumann entanglement entropy is [6]:

$$S(A) = |A| - n + \text{rank}_{\text{GF}(2)}(M_B), \quad (2)$$

where M_B is the $n \times 2|B|$ binary matrix formed by restricting the n stabilizer generators to the X - and Z -columns on subsystem B . The rank over $\text{GF}(2)$ —computed via Gaussian elimination with XOR arithmetic (see Appendix)—measures the dimension of the image of the stabilizer group under restriction to B : equivalently, it counts how many independent stabilizer constraints can be “seen” from B alone. The difference $|A| - n + \text{rank}(M_B)$ then equals the number of independent entangled Bell pairs across the $A|B$ cut, since each such pair contributes one unit of entropy and reduces the number of stabilizers fully visible from B .

We verify Eq. (2) against the Bell state $|\Phi^+\rangle$: stabilizers $\{X \otimes X, Z \otimes Z\}$ give $\text{rank}_{\text{GF}(2)}(M_B) = 2$ (with $B = \{\text{qubit } 1\}, n = 2, M_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$), yielding $S = 1 - 2 + 2 = 1$ ebit, as expected.

Phase Structure and Finite-Size Scaling

In the volume-law phase ($p < p_c$), the steady-state half-chain entropy satisfies $S(L/2) \propto L$. Li *et al.* [1] find that the volume-law phase also contains a subleading logarithmic correction: $S(A) = \alpha \ln |A| + s|A|$, where s is the volume-law coefficient and α captures universal fluctuations. In the area-law phase ($p > p_c$), $S(L/2) = O(1)$ independent of L . At the critical point, entanglement is purely logarithmic:

$$S(L/2) = \alpha(p_c) \ln(L/2) + \text{const}, \quad (3)$$

with $\alpha(p_c) \approx 1.6$ for random Clifford circuits at $L \gg 100$ [1]. Note that α is the raw log-scaling coefficient at this non-unitary critical point; its relation to CFT central charge c is discussed in Sec. .

Near the critical point, observables obey the finite-size scaling ansatz of Jian *et al.* [7]:

$$S(L, p) = f\left((p - p_c) \cdot L^{1/\nu}\right), \quad (4)$$

for a universal scaling function f and correlation length exponent ν . This is a simplified form: at p_c the entropy also contains the logarithmic term in Eq. (3), so the full scaling function near criticality reads $S(L, p) = \alpha \ln(L/2) + g((p - p_c)L^{1/\nu})$. Literature values are $p_c \approx 0.16$ and $\nu \approx 1.28$ – 1.3 [1, 8], obtained from system sizes up to $L \sim 512$.

METHODS

Circuit Architecture

The hybrid circuit is illustrated in Fig. 1. Each discrete time step t consists of two layers: (i) a brickwork layer of random two-qubit Clifford gates—even steps pair sites $(0, 1), (2, 3), \dots$, odd steps pair $(1, 2), (3, 4), \dots$, with unpaired boundary qubits receiving a random single-qubit Clifford; (ii) a measurement layer in which each qubit is independently measured in the Z basis with probability p , with outcomes discarded.

Random two-qubit Cliffords are drawn from the ensemble generated by the decomposition $C = (1Q) \cdot \text{CNOT}_{01} \cdot (1Q) \cdot \text{CNOT}_{10} \cdot (1Q) \cdot (1Q)$, where each $1Q$ denotes an independent uniformly random element of the verified 24-element single-qubit Clifford group. The single-qubit Clifford list is generated programmatically by enumerating all H/S sequences of length ≤ 6 , computing the Pauli-conjugation action of each via the stabilizer tableau, and deduplicating by action; an import-time assertion confirms exactly 24 distinct actions. This is a random entangling Clifford ensemble of the same general type used in stabilizer MIPT studies [1] and is expected to produce scrambling Clifford dynamics in the MIPT

universality class. The two-qubit gate ensemble is not verified as a uniform sampler over the full 11,520-element two-qubit Clifford group.

Simulation Protocol

We simulate system sizes $L \in \{8, 12, 16, 20, 24\}$ and measurement rates p at 26 equally spaced points in $[0.0, 0.5]$. For each (L, p) , we run 200 independent disorder realizations. Each realization begins in the product state $|0\rangle^{\otimes L}$ and applies $3L$ warmup layers at rate p (with measurements applied during warmup to correctly thermalize into the steady state of the hybrid Markov chain at rate p), followed by $4L$ main layers; the half-chain entropy is recorded after the final layer. We report the disorder average $\overline{S(L/2)}$ and standard error of the mean $\sigma/\sqrt{N_{\text{samples}}}$.

Computational Details

Entanglement entropy reduces to a $\text{GF}(2)$ rank computation via Eq. (2), executable in $O(n^3/64)$ bit-operations per sample with dense binary matrices. All gate operations are vectorized over tableau rows using NumPy array slicing, achieving $O(n)$ time per gate with no Python loop over rows. Total simulation: $\sim 26,000$ circuit runs across the (L, p) grid. Wall-clock time scales as L^3 (dominant cost: n^2 per measurement $\times O(L)$ steps): approximately 2–3 hours on a single CPU core (hardware-dependent estimate). Source code is available at github.com/agentjakey/miet-clifford, implemented from scratch without Stim or Qiskit.

RESULTS

Phase Transition

Figure 2 shows $\overline{S(L/2)}$ as a function of p for all five system sizes. The entropy is approximately proportional to L at $p = 0$ (volume-law coefficient $\alpha_{\text{vol}} = 0.995 \pm 0.017$, consistent with ~ 1 ebit per qubit), and saturates to $S = 0.105 \pm 0.022$ ebits at $p = 0.5$ for $L = 24$ (area law). The crossing of adjacent L pairs yields $p_{\text{cross}}^{\text{eff}} \in \{0.252, 0.211, 0.156, 0.192\}$ for increasing L pairs, with the largest- L effective crossing ($L = 20$ vs. $L = 24$) at $p_{\text{cross}}^{\text{eff}} \approx 0.192$. The crossing drift is not monotone in L , indicating that the crossing estimate should be treated as a finite-size diagnostic rather than a controlled extrapolation to the thermodynamic critical point. Averaging over all adjacent pairs gives the finite-size effective crossing estimate $p_c^{\text{eff}} = 0.203 \pm 0.017$. The reported ± 0.017 is a bootstrap statistical uncertainty only; the systematic finite-size bias—which shifts p_c^{eff} above the thermo-

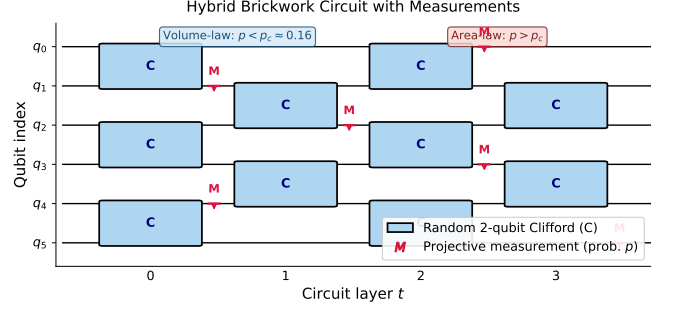


FIG. 1. Hybrid random Clifford circuit. Brickwork layers of random two-qubit Clifford gates (blue boxes labeled C) alternate with single-site measurements (red M symbols); each qubit is measured independently with probability p per layer, so the measurement locations shown are illustrative. Even layers pair qubits (0, 1), (2, 3), (4, 5); odd layers pair (1, 2), (3, 4). The competition between unitary scrambling and projective measurement drives the entanglement crossover at $p \approx p_c$.

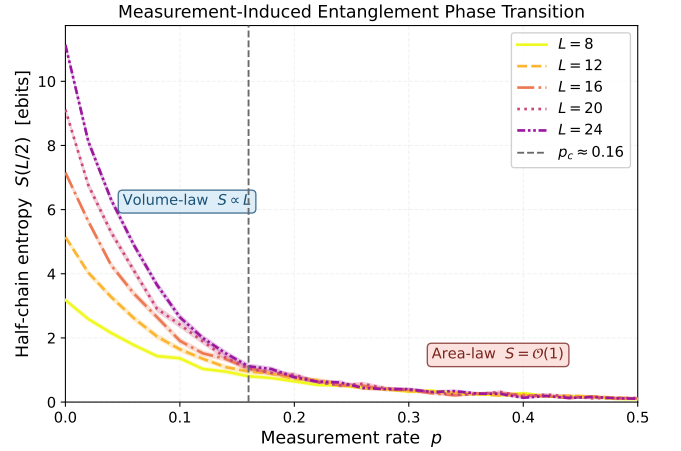


FIG. 2. Half-chain entanglement entropy $S(L/2)$ vs. measurement rate p for $L \in \{8, 12, 16, 20, 24\}$ (200 disorder realizations each; shaded bands: ± 1 standard error of the mean). The curves show finite-size signatures of the volume-law regime (S growing with L at low p) and the area-law regime (S approximately L -independent at high p); the thermodynamic transition is not fully resolved at these system sizes. Adjacent- L curve crossings yield a finite-size effective critical rate $p_c^{\text{eff}} \approx 0.20$ (statistical uncertainty ± 0.017), which lies above the thermodynamic value $p_c \approx 0.16$ [1] due to expected finite-size corrections.

dynamic $p_c \approx 0.16$ —is not reduced by collecting more samples at fixed L and is the dominant source of deviation from the literature value. This upward bias is a known feature of MIPT finite-size studies at $L \lesssim 50$ [8].

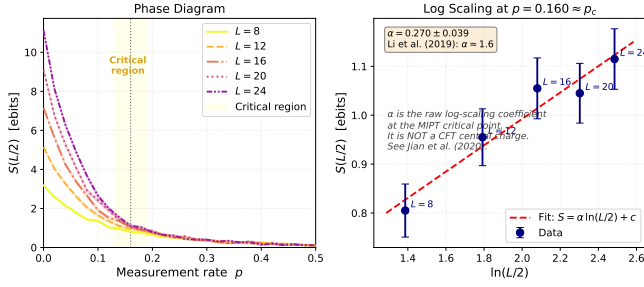


FIG. 3. (Left) Phase diagram with critical region $p \in [0.13, 0.19]$ shaded. (Right) $S(L/2)$ vs. $\ln(L/2)$ at $p = 0.16$ (grid point closest to the literature p_c). The linear fit (dashed red) yields the small-system fit coefficient $\alpha_{\text{eff}} = 0.27 \pm 0.07$, which is not quantitatively consistent with the asymptotic value $\alpha \approx 1.6$ [1] (established at $L \gg 100$); the suppression reflects off-critical evaluation and finite-size effects at $L \leq 24$ (see text). The figure provides evidence of qualitative critical-like logarithmic growth; it does not resolve the asymptotic exponent. The coefficient α is not a CFT central charge (see Sec.).

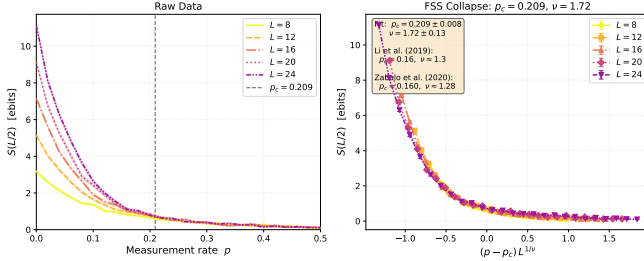


FIG. 4. Diagnostic finite-size scaling collapse under the simplified ansatz of Eq. (4) (without the critical logarithm; see text). (Left) Raw $S(L/2)$ vs. p for all five system sizes. (Right) Collapsed curves with finite-size effective parameters $p_c^{\text{eff}} = 0.209 \pm 0.008$, $\nu_{\text{eff}} = 1.72 \pm 0.10$ (bootstrap statistical uncertainty, 100 resamples). Because the raw entropy contains additive logarithmic corrections at p_c [Eq. (3)], the fitted p_c^{eff} and ν_{eff} should not be interpreted as precision thermodynamic estimates; the collapse is qualitative. Thermodynamic values: $p_c \approx 0.16$, $\nu \approx 1.28$ [8]. Quoted $\pm$ values are statistical only.

Logarithmic Scaling at the Critical Point

Figure 3 (right panel) shows $S(L/2)$ vs. $\ln(L/2)$ at $p = 0.16$, the grid point closest to the literature p_c . A linear fit over $L \in \{8, 12, 16, 20, 24\}$ gives slope $\alpha_{\text{eff}} = 0.270 \pm 0.074$. This value is *not* quantitatively consistent with the asymptotic literature value $\alpha \approx 1.6$ [1]. The fit is underpowered and strongly affected by two compounding effects: (i) at these system sizes the finite-size effective critical rate is $p_c^{\text{eff}} \approx 0.20$, so evaluating at $p = 0.16$ places the system in the volume-law side of the finite-size crossover rather than at true criticality—the logarithmic term is there, but so is a dominant linear-in- L contribution that suppresses the apparent slope; and (ii) even at

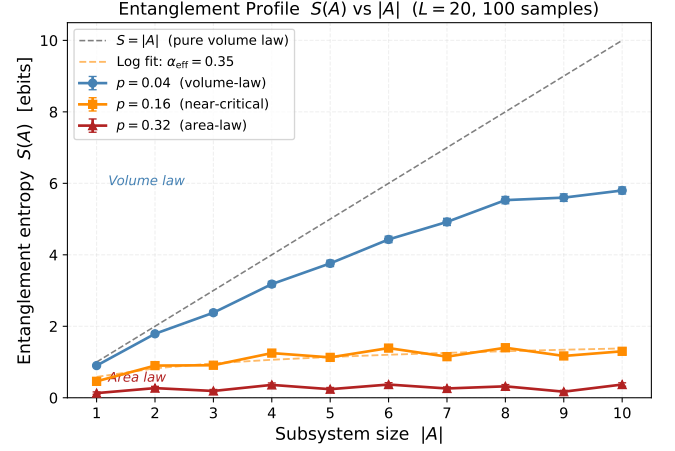


FIG. 5. Entanglement profile $S(A)$ vs. subsystem size $|A|$ for $L = 20$ (100 disorder realizations; error bars: ± 1 SEM). The contiguous subsystem $A = \{0, \dots, |A| - 1\}$. Three representative rates are shown: $p = 0.04$ (volume-law phase), $p = 0.16$ (close to the thermodynamic literature value, but below the finite-size effective crossing in our data), and $p = 0.32$ (area-law phase). The dashed black line $S = |A|$ marks the pure volume-law reference. In the volume-law phase the profile grows roughly linearly with $|A|$; in the area-law phase it is approximately flat; near p_c the growth is sublinear. The orange dashed curve shows a log fit $S = \alpha_{\text{eff}} \ln |A| + c$ to the $p = 0.16$ data ($\alpha_{\text{eff}} \approx 0.35$; the small value is expected at $L = 20$ as discussed in Sec.).

the true p_c , the asymptotic log-scaling regime of Eq. (3) requires $L \gg 100$ [8]. The data do show weak logarithmic growth of S with L , qualitatively compatible with critical-like behavior near the phase boundary. We report $\alpha_{\text{eff}} = 0.270$ as a finite-size effective coefficient and do not interpret it as a measurement of the thermodynamic exponent $\alpha(p_c)$.

Finite-Size Scaling Collapse

Figure 4 shows an FSS collapse under the simplified ansatz of Eq. (4). We note that this single-function form is an approximation: because $S(L, p_c)$ contains the logarithmic term in Eq. (3), the full scaling form near criticality reads $S(L, p) = \alpha \ln(L/2) + g((p - p_c)L^{1/\nu})$, where g is the universal scaling function. Fitting the simplified form without the logarithmic prefactor makes the collapse a qualitative diagnostic rather than a precision fit.

Minimizing the inter-curve mean-squared deviation over $L \in \{8, 12, 16, 20, 24\}$ yields finite-size effective parameters $p_c^{\text{eff}} = 0.209 \pm 0.008$ and $\nu_{\text{eff}} = 1.72 \pm 0.10$ (bootstrap statistical uncertainty, 100 resamples). Both estimates are substantially above the thermodynamic values ($p_c \approx 0.16$, $\nu \approx 1.28$ [8]). These deviations are expected systematic effects of finite-size corrections; the quoted bootstrap uncertainties quantify statistical precision only

TABLE I. Finite-size effective critical quantities from simulation vs. literature. System sizes $L \in \{8, 12, 16, 20, 24\}$, 200 samples per point. Reported $\pm$ values are bootstrap statistical errors only; systematic finite-size bias is the dominant source of deviation from thermodynamic values for p_c^{eff} , ν_{eff} , and α_{eff} .

Quantity	Finite-size eff. ($L \leq 24$)	Literature
p_c^{eff} (curve crossing)	0.203 ± 0.017	0.16 [1]
p_c^{eff} (FSS collapse)	0.209 ± 0.008	0.16 [1]
ν_{eff}	1.722 ± 0.104	1.28 [8]
α_{eff} (at $p = 0.16$)	0.270 ± 0.074	≈ 1.6 [1]
α_{vol} ($p = 0$)	0.995 ± 0.017	~ 1 (system dep.)
S_{∞} ($L = 24, p = 0.5$)	0.105 ± 0.022	$O(1)$

and do not account for this systematic bias, which is the dominant source of deviation at $L \leq 24$. Reliable extraction of thermodynamic exponents requires system sizes up to $L \sim \mathcal{O}(100)$ – $\mathcal{O}(500)$ [8], far beyond what is accessible here. At these system sizes, the five-curve collapse qualitatively captures the correct scaling structure, showing that the simulation qualitatively reproduces finite-size signatures associated with the MIPT.

Entanglement Profile vs. Subsystem Size

Figure 5 shows the disorder-averaged entanglement entropy $S(A)$ as a function of subsystem size $|A| = 1, 2, \dots, L/2$ for $L = 20$ at three representative measurement rates. In the volume-law phase ($p = 0.04 \ll p_c$), the profile grows roughly linearly with $|A|$: the best-fit slope is ≈ 0.55 ebits per qubit, consistent with $S(A) \propto |A|$ and supporting volume-law scaling beyond the half-chain entropy alone. In the area-law phase ($p = 0.32 \gg p_c$), the profile is approximately flat across subsystem sizes, with $S(A) \approx 0.13$ – 0.37 ebits over the full range $|A| = 1$ – 10 : the entropy does not grow appreciably with the subsystem size, as expected for a product-state-like phase. At $p = 0.16$ (close to the thermodynamic literature value, but below the finite-size effective crossing in our data), the profile is intermediate: sublinear growth, approximately consistent with $S(A) \sim \alpha \ln |A| + c$ with $\alpha_{\text{eff}} \approx 0.35$, reinforcing the near-critical interpretation from Sec. .

This profile analysis directly resolves the two phases at the level of individual subsystem sizes, strengthening the half-chain entropy evidence of Fig. 2: the volume-law and area-law behaviors are not an artifact of the specific bipartition $|A| = L/2$, but persist across all subsystem sizes. We note that at these system sizes ($L = 20$) the profile has not entered the asymptotic critical regime, and the slope in the volume-law phase reflects the finite-size steady-state rather than the true thermodynamic limit.

DISCUSSION

Quantum Error Correction Threshold Analogy

The MIPT is formally analogous to the quantum error correction (QEC) threshold theorem. In the area-law phase ($p > p_c$), measurements destroy quantum information faster than unitary scrambling can protect it: an adversary who monitors all measurement outcomes can, in principle, reconstruct the logical state from the syndrome. In the volume-law phase ($p < p_c$), scrambling delocalizes quantum information so thoroughly that even complete knowledge of measurement outcomes is insufficient to reconstruct the logical state [9]. The critical rate p_c is therefore closely related to the maximum error rate below which quantum information can in principle be preserved—a structural analog of the fault-tolerance threshold. This connection to QEC motivates practical interest in MIPT physics: understanding how p_c scales with circuit architecture has direct implications for the design of robust quantum error-correcting codes.

The Log-Scaling Exponent and the Replica Field Theory

At $p = p_c$, entanglement scales as $S(L/2) = \alpha(p_c) \ln(L/2) + \text{const}$ with $\alpha \approx 1.6$ [1] (established at $L \gg 100$). Appendix (Fig. 6) shows $\alpha_{\text{eff}}(p)$ computed by fitting $S(L/2) = \alpha_{\text{eff}} \ln(L/2) + c$ at each grid point p using all five system sizes. The coefficient α_{eff} peaks near the finite-size effective crossing $p_c^{\text{eff}} \approx 0.20$ and drops rapidly on both sides, consistent with the expectation that the log-scaling regime is only entered at or above p_c . The small values ($\alpha_{\text{eff}} \lesssim 0.5$) throughout confirm that these system sizes have not entered the asymptotic critical regime.

A standard unitary $(1+1)\text{D}$ CFT with central charge c would predict $\alpha = c/6$ for a system with open boundaries (Calabrese and Cardy [10]), which would give $c \approx 9.6$ —an unphysical value for any conventional CFT.

The reason is that the MIPT critical point is *not* a standard unitary CFT. Jian *et al.* [7]—including course instructor Prof. Yi-Zhuang You as a co-author—map the MIPT to a two-dimensional replica statistical mechanics model in which the $n \rightarrow 1$ replica limit generates an effective non-unitary field theory. The coefficient α characterizes the scaling dimension of the twist operator at this non-unitary fixed point and has no direct interpretation as a Virasoro central charge. Bao, Choi, and Altman [9] confirm this picture from a complementary random tensor network perspective. We therefore report α as a raw numerical exponent, explicitly avoiding the notation $c_{\text{eff}} = 6\alpha$ throughout this work.

Information Scrambling and the Correlation Length

Quantum scrambling is the process by which local information spreads under unitary evolution, becoming inaccessible to any local probe. In Clifford circuits, scrambling is quantified by the growth rate of the stabilizer generators’ spatial support. Near the critical point, the correlation length diverges as $\xi \sim |p - p_c|^{-\nu}$, signaling a renormalization group fixed point. The competition between unitary scrambling and measurement-induced localization drives this divergence; sufficiently close to p_c , the system cannot distinguish the volume-law and area-law phases on a finite chain. Out-of-time-order correlators (OTOCs) [1] provide the standard diagnostic for this competition, measuring how much a local operator spreads under evolution; their behavior near p_c encodes the same universality class as the entropy. The thermodynamic exponent $\nu \approx 1.3$ [8] implies that the critical window in measurement rate narrows as $\Delta p \sim L^{-1/\nu}$ with increasing system size. Our finite-size effective value $\nu_{\text{eff}} = 1.72 \pm 0.10$ is still converging toward this thermodynamic value and should not be interpreted as evidence for a distinct universality class.

Structural Analogies with Threshold Phenomena in Machine Learning

The MIPT belongs to a broader class of *threshold phenomena in complex information-processing systems*: below a critical disturbance rate, structured information survives; above it, it irreversibly collapses. We note two speculative structural analogies, emphasizing that these are heuristic comparisons with no known quantitative mapping, not formal equivalences.

Grokking. In neural network training, “grokking” [11] refers to a sharp generalization transition—delayed memorization followed by sudden algorithmic compression—that emerges after prolonged training on small datasets. Both grokking and the MIPT exhibit a sharp crossover between two qualitatively distinct regimes as a single control parameter crosses a threshold; in both cases, the transition is accompanied by a structural reorganization of internal representations rather than a smooth interpolation. The analogy is qualitative only.

Robustness of learned representations under optimization pressure. Whether safety-relevant internal representations survive continued fine-tuning is structurally analogous to asking whether volume-law entanglement survives increasing measurement rate. Near the critical point, the tuning window in p narrows as $\Delta p \sim L^{-1/\nu}$ as system size grows, suggesting that analogous robustness thresholds in ML systems may sharpen with model scale. We emphasize this comparison is heuristic: the MIPT exponent ν has no known quantitative relationship to learning dynamics. Information-delocalization measures—

effective representation rank, participation ratio in the basis of internal representations—may nonetheless serve as diagnostic tools for studying representation robustness in mechanistic interpretability research, motivated by this analogy.

These analogies are speculative and are offered as motivation for future cross-disciplinary investigation, not as formal claims.

Limitations of the Finite-Size Analysis

The system sizes $L \in \{8, 12, 16, 20, 24\}$ are sufficient to exhibit the qualitative finite-size signatures of the MIPT—the crossover between volume-law and area-law entropy, the drift of adjacent- L crossings, and an approximately logarithmic growth of S near the critical window—but they are too small for reliable extraction of thermodynamic critical exponents. Precision MIPT studies reach $L \sim 512$ [8]; at $L \leq 24$, the system has not entered the asymptotic scaling regime for any of the exponents considered here.

The bootstrap uncertainties reported throughout— $\sigma(p_c^{\text{eff}}) \approx 0.008$, $\sigma(\nu_{\text{eff}}) \approx 0.10$, $\sigma(\alpha_{\text{eff}}) \approx 0.07$ —quantify only the statistical precision of the estimators given the available finite-size data. They do not capture any of the systematic uncertainties that dominate the deviation from thermodynamic values: finite-size corrections to scaling (the leading source), sensitivity to the choice of gate ensemble, finite circuit depth relative to the correlation time, choice of scaling ansatz (simplified vs. full critical form), the fitting window over which collapse quality is evaluated, and the choice of observable (here, half-chain entropy at a fixed bipartition). These systematic contributions cannot be reduced by collecting more disorder realizations at the same system sizes.

The FSS collapse of Fig. 4 is diagnostic rather than quantitative. Because the critical entropy contains an additive logarithmic term [Eq. (3)], the simplified single-function ansatz of Eq. (4) is approximate near p_c ; the fitted parameters absorb the missing logarithm into an effective p_c^{eff} and ν_{eff} that systematically differ from thermodynamic values. The collapse demonstrates that the simulation reproduces the correct qualitative scaling structure; it does not constitute a precision measurement of the universality class exponents.

The α_{eff} fit is additionally underpowered. It is computed from five data points evaluated at the single grid value $p = 0.16$, which lies below the finite-size effective critical rate $p_c^{\text{eff}} \approx 0.20$. A small shift in the evaluation point changes the fitted slope substantially, because the entropy at these sizes has not entered the asymptotic critical log-scaling regime. The resulting $\alpha_{\text{eff}} = 0.270 \pm 0.074$ is a finite-size effective coefficient, not an estimate of the thermodynamic value $\alpha(p_c) \approx 1.6$.

This project is best understood as a from-scratch nu-

merical reproduction of the qualitative phenomenology of the MIPT: the crossover, the logarithmic signature, and the finite-size scaling structure are all visible and physically interpretable. The implementation is verified by seven independent physics checks and produces results consistent with the qualitative predictions of the field. It is not a precision numerical study.

CONCLUSION

We implemented the Aaronson-Gottesman stabilizer tableau from scratch and used it to study measurement-induced phase transitions in one-dimensional random Clifford circuits. Simulating $L \in \{8, 12, 16, 20, 24\}$ with 200 disorder realizations per point, we reproduced the qualitative MIPT phenomenology: a crossover from volume-law ($S \propto L$) to area-law ($S = O(1)$) entanglement as the measurement rate increases past a finite-size effective critical rate $p_c^{\text{eff}} \approx 0.20$. We additionally computed the full entanglement profile $S(A)$ vs. $|A|$ for $L = 20$ at three representative rates, directly confirming the linear volume-law growth ($S \approx 0.55|A|$ at $p = 0.04$) and the flat area-law plateau ($S \lesssim 0.4$ ebits across all subsystem sizes at $p = 0.32$). All system sizes studied are too small for reliable thermodynamic exponent extraction: the extracted finite-size effective parameters ($p_c^{\text{eff}} \approx 0.20$, $\nu_{\text{eff}} \approx 1.72$, $\alpha_{\text{eff}} \approx 0.27$) are substantially affected by finite-size corrections and are not quantitatively consistent with thermodynamic literature values [1, 8]; the quoted bootstrap uncertainties are statistical only and do not reduce the dominant systematic finite-size bias. The volume-law coefficient $\alpha_{\text{vol}} = 0.995 \pm 0.017$ and the area-law saturation $S_{\infty} = 0.105 \pm 0.022$ ebits are physically interpretable and independently verified; the simulator passes 57 automated tests including exact Bell-state and GHZ entropy checks, gate conjugation identities, stabilizer commutation invariants, measurement repeatability, and a demonstration that GF(2) rank diverges from real-valued rank on a constructed counter-example.

We discussed the formal analogy between p_c and the quantum error correction threshold, the replica-field-theory interpretation of the log-scaling exponent α (which is *not* a CFT central charge [7]), and speculative structural analogies with threshold phenomena in machine learning.

Future directions that would make this a precision numerical study include verifying whether the two-qubit gate decomposition uniformly samples the full 11,520-element two-qubit Clifford group (single-qubit uniformity is now verified; two-qubit uniformity is not), extending to larger system sizes ($L \sim 100$ –500) where the asymptotic log-scaling and FSS regimes are accessible, implementing OTOC diagnostics to directly measure scrambling, and studying the entanglement profile at criticality to more precisely extract $\alpha(p_c)$ as a function of $|A|$.

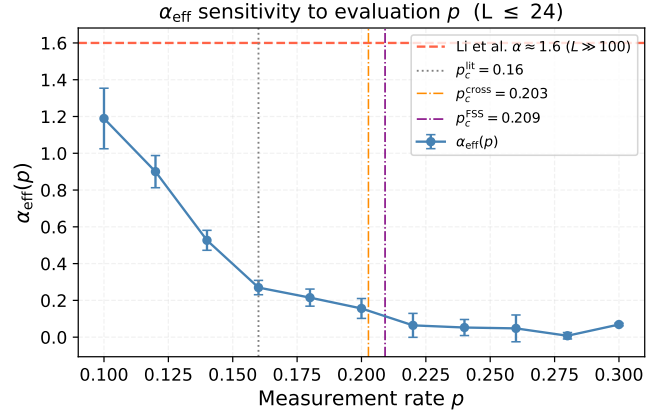


FIG. 6. Small-system log-scaling coefficient $\alpha_{\text{eff}}(p)$ obtained by fitting $S(L/2) = \alpha_{\text{eff}} \ln(L/2) + c$ to the five system sizes $L \in \{8, 12, 16, 20, 24\}$ at each measurement rate p . Error bars are OLS standard errors. The orange dashed line marks the thermodynamic value $\alpha \approx 1.6$ [1] (established at $L \gg 100$); the grey dotted line marks $p_c^{\text{lit}} = 0.16$. The coefficient peaks near the finite-size effective crossing $p_c^{\text{eff}} \approx 0.20$ and is everywhere well below 1.6, confirming that the asymptotic critical log-scaling regime is not reached at $L \leq 24$.

The author thanks Prof. Yi-Zhuang You for lecture material motivating this investigation, and acknowledges Ref. [7] (co-authored by Prof. You) as a foundational analytical reference for the replica field theory of the MIPT. Simulations used Claude Code (Anthropic) for AI-assisted implementation of the stabilizer tableau and circuit infrastructure.

Log-Scaling Coefficient vs. Measurement Rate

Figure 6 shows $\alpha_{\text{eff}}(p)$ computed by fitting $S(L/2) = \alpha_{\text{eff}} \ln(L/2) + c$ at each grid point p using all five system sizes. This supplemental figure illustrates the sensitivity of the log-scaling coefficient to the evaluation point discussed in Sec. .

Stabilizer Tableau Gate Update Rules

Following Aaronson and Gottesman [5] (Table 1), the gate update rules are applied simultaneously to all rows $h \in \{0, \dots, 2n - 1\}$ of the tableau. Let $x_{h,j}$ and $z_{h,j}$ denote the X - and Z -bits for qubit j in row h , and r_h the phase bit.

Hadamard on qubit a . The X - and Z -bits are swapped; the phase flips whenever both were 1:

$$\begin{aligned} r_h \oplus &= x_{h,a}^{\text{old}} \cdot z_{h,a}^{\text{old}}, \\ x_{h,a} &\leftarrow z_{h,a}^{\text{old}}, \quad z_{h,a} \leftarrow x_{h,a}^{\text{old}}. \end{aligned} \quad (5)$$

The copy of $x_{h,a}^{\text{old}}$ must be saved before any modification.

Phase (S) gate on qubit a . The phase flips when $x_{h,a} = 1$ and the old $z_{h,a} = 1$ (i.e., the generator has Y on qubit a); then $z_{h,a}$ is toggled:

$$\begin{aligned} r_h \oplus &= x_{h,a} \cdot z_{h,a}^{\text{old}}, \\ z_{h,a} \oplus &= x_{h,a}. \end{aligned} \quad (6)$$

One can verify: $S|Y\rangle \rightarrow -|X\rangle$, i.e., $Y \mapsto -X$, consistent with $SY S^\dagger = -X$.

CNOT with control a , target b .

$$\begin{aligned} r_h \oplus &= x_{h,a} \cdot z_{h,b} \cdot (x_{h,b} \oplus z_{h,a} \oplus 1), \\ x_{h,b} \oplus &= x_{h,a}, \\ z_{h,a} \oplus &= z_{h,b}. \end{aligned} \quad (7)$$

All three updates use copies of the pre-update columns; the order of the last two may be exchanged freely.

Rowsum phase function. The measurement algorithm requires computing the product of two Pauli operators. The phase contribution is [5]:

$$g(x_1, z_1, x_2, z_2) = \begin{cases} z_2 - x_2 & x_1 = 1, z_1 = 1, \\ z_2(2x_2 - 1) & x_1 = 1, z_1 = 0, \\ x_2(1 - 2z_2) & x_1 = 0, z_1 = 1, \\ 0 & x_1 = 0, z_1 = 0, \end{cases} \quad (8)$$

and the new phase bit of row h after adding row i is:

$$r_h \leftarrow \left\lfloor \frac{1}{2} \left(2r_h + 2r_i + \sum_j g_{ij} \right) \bmod 4 \right\rfloor, \quad (9)$$

where $g_{ij} \equiv g(x_{i,j}, z_{i,j}, x_{h,j}, z_{h,j})$ and the sum runs over $j = 0, \dots, n-1$. The sum runs over all n qubits and must be computed in ordinary (mod-4) integer arithmetic—not mod-2.

GF(2) Rank Algorithm

Entanglement entropy requires computing $\text{rank}_{\text{GF}(2)}(M_B)$, the rank of the binary matrix M_B over the field with two elements. **numpy.linalg.matrix_rank operates over $\mathbb{R}$ and cannot be used here:** floating-point arithmetic does not respect GF(2) arithmetic, and it returns incorrect ranks for binary matrices with linearly dependent rows that sum to zero modulo 2.

The correct algorithm is Gaussian elimination with XOR operations:

1. Copy M_B to a working matrix M of dtype `int8`.
2. Initialize `rank = 0` and `pivot_row = 0`.
3. For each column $c = 0, 1, \dots, 2|B| - 1$:
 - (a) Find the first row $r \geq \text{pivot_row}$ with $M[r, c] = 1$. If none, skip to the next column.
 - (b) Swap rows r and `pivot_row`.

- (c) For every other row $r' \neq \text{pivot_row}$ with $M[r', c] = 1$: set $M[r'] \oplus= M[\text{pivot_row}]$ (XOR, element-wise).

- (d) Increment `rank` and `pivot_row`.

4. Return `rank`.

XOR ($\oplus$) implements addition in GF(2) without modular reduction overhead. The algorithm runs in $O(n \cdot |B|^2)$ time, which is negligible compared to the $O(n^2)$ cost of a single measurement operation.

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